TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Assessment of Quality of Fabric

Visual inspection on the frame and the arithmetic of acceptance — the four-point system, points per hundred square yards, and the grading that sorts first quality from seconds.

TE-3202 · QUALITY EVALUATION OF FABRIC AND APPAREL SESSIONAL · EXPERIMENT 14 · LEVEL 3 TERM 2

1. Aim and theory

Aim: to assess the quality of a fabric roll by visual inspection on an inspection frame and to grade it by the four-point system (ASTM D5430 — grading, classifying and scoring of fabrics), calculating penalty points and points per 100 square yards, judging the roll against a buyer limit, and outlining the ten-point system and acceptance sampling of rolls.

1.1 Inspection: seeing the fabric as the customer will

Every laboratory property in this course is measured on samples; fabric inspection is different — it examines the product itself, at full width and production length, for the visible faults that sampling plans cannot map: slubs, broken ends and picks, holes, stains, barré, shade variation, misprints, skew and bow. The fabric passes over an **inspection frame** — an inclined, evenly lit table — at an agreed speed slow enough for the inspector to see defects at the fabric's true viewing distance and lighting. Agreed is the operative word: illumination, viewing distance, frame angle and running speed are written into the inspection agreement between mill and buyer, because a fault that is invisible at one speed and glaring at another otherwise becomes a dispute instead of a measurement. The inspector marks each defect at the selvedge (ticket or chalk position), records its type and size on the inspection sheet, and cuts or leaves defects according to the agreement. Judgment is trained and audited: inspectors are checked against reference rolls, since the system assumes a defect seen is a defect scored the same way by every inspector.

1.2 The four-point system (ASTM D5430)

The four-point system converts what the inspector sees into one comparable number. Each defect is scored by its size — for running defects, its length in the direction of its greatest extent:

DEFECT SIZE	PENALTY POINTS
Up to 3 in	1 point

Over 3 in, up to 6 in	2 points
Over 6 in, up to 9 in	3 points
Over 9 in	4 points
Hole or opening over 1 in	4 points

Three discipline rules keep the score honest, and the student should state the core set exactly. No linear yard of fabric may be assigned more than 4 points in total, however crowded with defects that yard is — the cap stops one catastrophic yard from swamping an otherwise sound roll, and it forces long running defects to be scored yard by yard rather than as one endless penalty. A defect continuing through more than one yard is scored in each yard it enters, subject to the same cap. Holes are scored by size as defects, with a hole over 1 in taking the maximum 4 points. Because rolls differ in length and width, raw totals are never compared: the score is normalised to the area inspected:

$$\text{Points per 100 yd}^2 = \text{total penalty points} \times 100 / \text{fabric area inspected (yd}^2\text{)}$$

$$\text{Fabric area (yd}^2\text{)} = \text{roll length (yd)} \times \text{width (in)} / 36$$

The buyer sets the acceptance limit for points per 100 square yards — commonly in the range of 20–40 depending on fabric class and end use, with the figure in this exercise given as 28. At or under the limit the roll is **first quality**; above it, the roll is downgraded to seconds, price-reduced, cut around at a loss, or rejected, as the contract provides. Note what the number does not do: it weights a 2-inch slub and a 2-inch oil spot equally, and it says nothing about where defects fall across the width. Mills therefore also record defect type and position, and some faults (a repeating temple mark, a shade change across the roll) fail a roll on sight regardless of arithmetic.

1.3 The ten-point system in outline

The older ten-point system, still met in some trades, scores warp and weft defects separately on a finer scale: warp defects are penalised from 1 point for a fault up to 1 inch up to 10 points for a fault exceeding a yard, with weft defects on a related scale, and the total is expressed per linear yard rather than per area. Its finer granularity rewards careful counting, but its warp/weft bookkeeping and per-yard basis made cross-roll comparison harder, which is why most modern apparel trade has settled on the four-point, area-based method taught here. A graduate should recognise a ten-point report, know that its numbers are not interchangeable with four-point numbers, and never convert between systems by a rule of thumb.

1.4 Sampling of rolls: AQL in outline

No buyer inspects every roll of a large delivery, and no honest mill asks them to trust luck. Rolls are selected by an acceptance sampling plan on AQL (acceptable quality limit) principles: the lot size

determines the sample size, the sampled rolls are inspected and graded as above, and the lot is accepted or rejected against pre-agreed accept/reject numbers of defective rolls (or of points, where the contract grades lots that way). The plan's mathematics balances two risks — the producer's risk that a good lot is rejected by an unlucky sample, and the consumer's risk that a bad lot passes through one — and both parties sign the plan before the goods are made, not after a dispute begins. Inspection of the sampled rolls is full-length inspection: sampling chooses which rolls, never which metres.

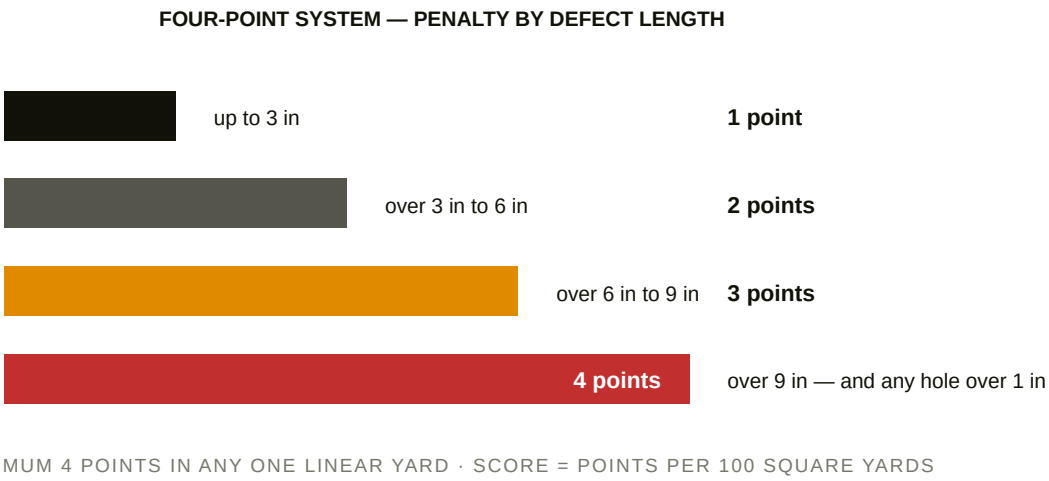


Figure 1 — The four-point penalty scale drawn to length: each step of defect size adds one point, capped at 4 points for defects over 9 inches, for holes over 1 inch, and for any single linear yard however defective.

2. Apparatus and procedure

- **Apparatus:** inspection frame with agreed lighting and controlled, steady running speed; measuring rule or tape (inches and yards); selvedge marking tickets or chalk; inspection record sheet with defect codes; calculator; the roll mounted so fabric draws off without tension distortion.
- **Set-up:** confirm the inspection agreement before starting — lighting, speed, viewing distance, which defects count, and the acceptance limit. Condition of work: fabric inspected at rest width, roll identity and nominal length and width recorded at the head of the sheet.
- **Inspection:** run the full roll at the agreed speed with the inspector at the standard viewing position. At each defect, stop or mark as the agreement directs: identify the type, measure its greatest extent (and note a hole's size separately), assign 1–4 points by the scale, check the 4-points-per-linear-yard cap for the yard in which it falls, mark the selvedge at that position, and enter type, position and points on the record.
- **Running and full-width faults:** score a continuing defect in each yard it enters (subject to the cap) and record faults that span the width or repeat at intervals by their own agreement rules — never let a long fault be entered once and forgotten, and never let it be entered yard by yard beyond the cap either.

- **Calculation:** sum the penalty points; measure or verify the actual inspected length and width; compute the area in square yards and the points per 100 square yards. State the verdict against the limit and the disposition of the roll (first quality, seconds, or hold for buyer decision), with the major defect types summarised for the mill's corrective action.

3. Worked calculation

Roll inspected: length 120 yd, width 60 in. Defects found over the full roll:

DEFECT CLASS	COUNT	POINTS EACH	POINTS
Defects up to 3 in	7	1	7 × 1 = 7
Defects over 3 in to 6 in	6	2	6 × 2 = 12
Defects over 6 in to 9 in	5	3	5 × 3 = 15
Defects over 9 in	2	4	2 × 4 = 8
Hole over 1 in	1	4	1 × 4 = 4
Total	**21 defects**	—	**46 points**

$$\text{Fabric area} = 120 \text{ yd} \times (60 / 36) \text{ yd} = 120 \times 1.667 = 200 \text{ yd}^2 \text{ (exactly: } 120 \times 60 / 36 = 200 \text{ yd}^2\text{)}$$

$$\text{Points per } 100 \text{ yd}^2 = 46 \times 100 / 200 = 23.0 \text{ points per } 100 \text{ yd}^2$$

Against the buyer's limit of 28 points per 100 square yards, 23.0 passes with margin: the roll is graded **first quality**. Two features of the tally deserve the mill's attention even in a pass. The largest single contribution (15 of 46 points) comes from the 6–9 inch class — a small number of substantial defects, worth tracing to their machine cause — while the single hole, though only 4 points, is the defect type most likely to reach a garment panel and must be flagged at its selvedge position so cutting can plan around it. The score accepts the roll; the defect record improves the next one.

RESULT

Total penalty 46 points on 200 yd² = 23.0 points per 100 yd². Buyer limit 28 → first quality, pass. Largest contribution: 6–9 in defects (15 points); the one hole over 1 in is marked for the cutting room despite the comfortable overall score.

4. Precautions

- Inspect under the agreed lighting, speed and viewing distance only, and recalibrate inspectors against reference rolls. Changing the conditions changes the visible defect population and quietly rewrites the standard.
- Measure each defect by its greatest extent and apply the class boundaries exactly — a defect of exactly 3 in scores 1 point; one just over 3 in scores 2. Boundary generosity, repeated over a roll, is grade inflation.
- Apply the 4-points-per-linear-yard cap in both directions of honesty: never score a yard above 4, and never spread one yard's defects into neighbouring yards to evade the cap's intent or to inflate a claim.
- Verify the roll's actual length and width for the area calculation. Points per 100 square yards computed on the ticket length of a short roll flatters the mill and cheats the comparison.
- Mark every defect at the selvedge as it is scored, including in a passing roll. The grade decides acceptance; the marks decide whether the cutting room can use the accepted cloth.
- Record defect types faithfully alongside the points. A report of "23 points" alone gives production nothing to fix; the same report naming its 6–9 inch class sends maintenance to the right machine.

Practice (viva preparation)

1. State the four-point penalties for defects of 2 in, 5 in, 8 in and 12 in, and for a hole of 1.5 in.
2. Why is no linear yard ever scored above 4 points?
3. A roll is 120 yd long and 60 in wide. Find its area in square yards.
4. The roll of Question 3 carries 46 penalty points. Find its points per 100 square yards and judge it against a limit of 28.
5. Why are scores expressed per 100 square yards rather than as raw totals or per linear yard?
6. How does the ten-point system differ from the four-point system, in outline?
7. What does AQL sampling decide, and what does it never decide?

Answer key

1. 2 in → 1 point (up to 3 in). 5 in → 2 points (over 3 to 6 in). 8 in → 3 points (over 6 to 9 in). 12 in → 4 points (over 9 in). A hole of 1.5 in → 4 points, because any hole over 1 in takes the maximum penalty.
2. The cap keeps the score a measure of the roll as a whole. Without it, one short length of catastrophic fabric — a smashed pick running through a yard crowded with secondary faults — could contribute dozens of points and condemn a roll whose remaining hundred yards are clean, or dominate negotiations out of proportion to the usable cloth lost. Long defects are still scored — in each yard they enter — but each yard's contribution is bounded at 4.

3. Area = length × width in yards = 120 × (60/36) = 120 × 60 / 36 = 200 yd².
4. Points per 100 yd² = 46 × 100 / 200 = 23.0. Against a limit of 28, 23.0 is inside the limit — the roll is first quality and is accepted (with its defects marked and recorded as usual).
5. Rolls differ in both length and width, so a raw total rewards short, narrow rolls and punishes long, wide ones for containing more cloth, not worse cloth. Per linear yard removes the length bias but keeps the width bias. Area — per 100 square yards — is the only basis on which any two rolls, and any two deliveries, carry comparable scores.
6. The ten-point system scores warp and weft defects separately on a scale running to 10 points for the largest faults and expresses the result per linear yard; the four-point system uses a single 1–4 scale for all defects, caps each yard at 4 points, and expresses the result per 100 square yards. The systems' numbers are not interchangeable and are never converted into each other.
7. AQL sampling decides which rolls of a lot are inspected and, from the agreed accept/reject numbers applied to those rolls, whether the lot is accepted or rejected, balancing the producer's and consumer's risks. It never decides which parts of a chosen roll are inspected — a sampled roll is inspected for its full length.
